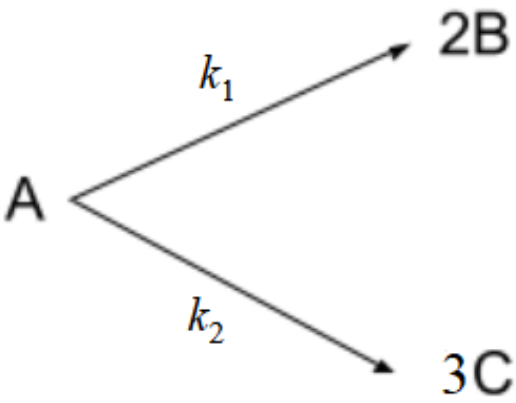
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Consider the following parallel reactions being given by A $= 1.386 \times 10^2$ hours), each path being first order.



If the distribution of B in the Product mixture is 50%, the partial half life of A for converison into B is

(A) $231\ \text{h}$

(B) $131\ \text{h}$

(C) $115.5\ \text{h}$

(D) $31\ \text{h}$

Q.2

A reactant (A) forms two products:

$\text{A} \xrightarrow{k_1} \text{B}$; Activation energy E_{a1}

$\text{A} \xrightarrow{k_2} \text{C}$; Activation energy E_{a2}

If $E_{\text{a2}} = 2E_{\text{a1}}$, then k_1 and k_2 are related as

(A) $k_1 = 2k_2 \text{e}^{E_{\text{a2}}/RT}$

(B) $k_1 = k_2 \text{e}^{E_{\text{a1}}/RT}$

(C) $k_2 = k_1 \text{e}^{E_{\text{a2}}/RT}$

(D) $k_1 = Ak_2 \text{e}^{E_{\text{a}}/RT}$

Q.3

For the first order reaction $\text{A} \rightarrow \text{B} + \text{C}$ is carried out at 27°C. If $3.8 \times 10^{-16}\%$ of the reactant molecules exists in the activated state, the E_e (activation energy) of the reaction is:

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(A) $12\ \text{kJ/mol}$

(B) $831.4\ \text{kJ/mol}$

(C) $100\ \text{kJ/mol}$

(D) $88.57\ \text{kJ/mol}$

Q.4

A catalyst lowers the activation energy of a certain reaction from 83.314 to $75\ \text{kJ mol}^{-1}$ at $500\ \text{K}$. What will be the rate of reaction as compared to uncatalysed reaction? Assume other things being equal.

(A) Double

(B) 28 times

(C) 7.38 times

(D) 7.38×10^3 times

Q.5

Rate of a reaction can be expressed by Arrhenius equation as: $k = Ae^{-E_a/RT}$
In this equation, E_a represents

(A) the total energy of the reacting molecules at a temperature, T

(B) the fraction of molecules with energy greater than the activation energy of the reaction

(C) the energy below which all the colliding molecules will react

(D) the energy below which colliding molecules will not react

Q.6

The reason for almost doubling the rate of reaction on increasing the temperature of the reaction system by 10°C is

(A) The value of threshold energy increases

(B) Collision frequency increases

(C) The fraction of the molecule having energy equal to threshold energy or more increases

(D) Activation energy decreases

Q.7

In Arrhenius plot, intercept is equal to

(A) $\frac{-\text{E}_\text{a}}{\text{R}}$

(B) $\ln \text{A}$

(C) $\ln \text{K}$

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(D) $\log_{10} A$

Q.8

The slope in Arrhenius plot, is equal to:

(A) $-\frac{E_a}{2.303R}$

(B) $\frac{E_a}{R}$

(C) $-\frac{R}{2.303E_a}$

(D) None of these

Q.9

When a biochemical reaction is carried out in laboratory in the absence of enzyme then rate of reaction obtained is 10^{-6} times, then activation energy of reaction in the Presence of enzyme is

(A) $\frac{6}{\mathrm{RT}}$

(B) Different from $\mathrm{E_a}$ obtained in laboratory

(C) P is required

(D) Can't say anything

Q.10

The energies of activation for forward and reverse reactions for $\mathrm{A_2 + B_2 \rightleftharpoons 2AB}$ are $180\ \mathrm{kJ\ mol^{-1}}$ and $200\ \mathrm{kJ\ mol^{-1}}$ respectively. The presence of a catalyst lowers the activation energy of both (forward and reverse) reactions by $100\mathrm{k.\ mol^{-1}}$. The enthalpy change of the reaction $\left(\mathrm{A_2 + B_2 \rightarrow 2AB}\right)$ in the presence of a catalyst will be (in $\mathrm{kJmol^{-1}}$) _____

(A) -20

(B) -15

(C) 20

(D) 15

Q.11

Rate constant $k = 1.2 \times 10^3\ \mathrm{mol^{-}1\ L\ s^{-1}}$ and $\mathrm{E_a} = 2.0 \times 10^2\ \mathrm{kJ\ mol^{-1}}$. When $\mathrm{T} \rightarrow \infty$

(A) $\mathrm{A} = 2.0 \times 10^2\ \mathrm{kJ\ mol^{-1}}$

(B) $\mathrm{A} = 1.2 \times 10^3\ \mathrm{mol^{-1}\ L\ s^{-1}}$

(C) $\mathrm{A} = 1.2 \times 10^3\ \mathrm{mol\ L^{-1}\ s^{-1}}$

(D) $\mathrm{A} = 2.4 \times 10^3\ \mathrm{kJ\ mol^{-1}\ s^{-1}}$

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Q.12

A catalyst lowers the activation enegy of a reaction from $20\ \text{kJ mol}^{-1}$ to $10\ \text{kJ mol}^{-1}$. The temperature at which the uncatalyzed reaction will have the same rate as that of the catalyzed a 27°C is

(A) -123°C

(B) 327°C

(C) 32.7°C

(D) $+23^{\circ}\text{C}$

Q.13

The activation energies of the forward and backward reactions in the case of a chemical reaction are 30.5 and $45.4\ \text{kJ/mol}$ respectively. The reaction is :

(A) exothermic

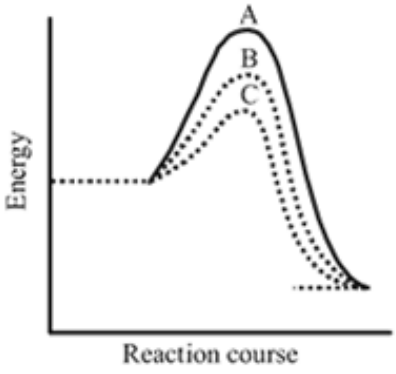
(B) endothermic

(C) neither exothermic nor endothermic

(D) independent of temperature

Q.14

A homogeneous catalytic reaction takes place through the three alternative plots A, B and C shown in the given figure. Which one of the following indicates the relative ease with which the reaction can take place?



(A) $\text{A} > \text{B} > \text{C}$

(B) $\text{C} > \text{B} > \text{A}$

(C) $\text{A} > \text{C} > \text{B}$

(D) $\text{A} = \text{B} = \text{C}$

Q.15

For an exothermic reaction, the energy of activation of the reactants is

(A) equal to the energy of activation of products

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(B) less than the energy of activation of products

(C) greater than the energy of activation of products

(D) sometimes greater and sometimes less than that of the products

Q.16

When a biochemical reaction is carried out in laboratory in the absence of enzyme then rate of reaction obtained is 10^{-6} times, then activation energy of reaction in the presence of enzyme is

(A) $\frac{6}{\mathrm{RT}}$

(B) different from $\mathrm{E_a}$ obtained in laboratory

(C) P is required

(D) can't say anything

Q.17

In a reaction at $27^{\circ}\mathrm{C}, 10^{-3}\%$ reactant molecules manage to cross over the barrier of transition state. The energy of these molecules in excess of the average value will be $\left(R=2 \mathrm{calK}^{-1} \mathrm{~mol}^{-1}\right)$:

(A) $6.91 \mathrm{kcalmol}^{-1}$

(B) $3.00 \mathrm{kcalmol}^{-1}$

(C) $4.15 \mathrm{kcalmol}^{-1}$

(D) $5.10 \mathrm{kcalmol}^{-1}$

Q.18

The activation energy for a simple chemical reaction $\mathrm{A} \rightarrow \mathrm{B}$ is E_a in forward direction. The activation energy for reverse reaction

(A) is always double of E_a

(B) is negative of E_a

(C) is always less than E_a

(D) can be less than or more than E_a

Q.19

Consider the consecutive reactions :

$$A \xrightarrow{k=2\times10^{-5}\ \mathrm{s}^{-1}} B \xrightarrow{k=8\times10^{-6}\ \mathrm{s}^{-1}} C \xrightarrow{k=3\times10^{-3}\ \mathrm{s}^{-1}} D$$

The rate determining step of the reaction is :

(A) $A \rightarrow B$

(B) $C \rightarrow D$

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(C) $B \rightarrow C$

(D) $A \rightarrow D$

Q.20

The rate constant k_1 and k_2 for two different reactions are $10^{16}e^{-2000/T}$ and $10^{15}e^{-1000/T}$ respectively. The temperature at which $k_1 = k_2$ is :-

(A) $2000\ \text{K}$

(B) $\frac{1000}{2.303}\ \text{K}$

(C) $1000\ \text{K}$

(D) $\frac{2000}{2.303}\ \text{K}$

Answers & Solutions

Q.1 Answer:

$231\ \text{h}$

Solution:

$\frac{2k_1}{2k_1+3k_2} = 0.5$ and

$k_1 + k_2 = \frac{0.693}{1.386\times10^2} = 5\times10^{-3}h^{-1}$

Solving $\frac{k_1}{k_2} = \frac{2}{3}$ **and** $k_1 = 2\times10^{-3}h^{-1}$ **and**

$k_2 = 3\times10^{-3}h^{-1}$

$t_{1/2(A\rightarrow B)} = \frac{0.693}{k_2} = \frac{0.693}{3\times10^{-3}} = 231h$

Q.2 Answer:

$k_1 = Ak_2e^{E_a/RT}$

Solution:

We have $k_1 = A_1e^{-E_{a1}/RT}$ **and** $k_2 = A_2e^{-E_{a2}/RT} = A_2e^{-2E_{a1}/RT}$

Hence $\frac{k_1}{k_2} = \frac{A_1}{A_2}\frac{e^{-E_{al}/RT}}{e^{-2E_{a1}/RT}} = \frac{A_1}{A_2}e^{E_{al}/RT}$ **or** $k_1 = \left(\frac{A_1}{A_2}\right)k_2e^{E_{a1}/RT} = Ak_2e^{E_{al}/RT}$

Q.3 Answer:

$100\ \text{kJ/mol}$

Solution:

$e^{-\frac{E_a}{RT}} = 3.8\times-\frac{10^{-16}}{100}$

$-\frac{E_a}{RT} = \ln 3.8\times10^{-18}$

$E_a = 100\ \text{kJ/mol}$

Q.4 Answer:

7.38 times

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Solution:

$\frac{k_2}{k_1} = \frac{Ae^{-Ea_2/RT}}{Ae^{-E_{a_1}/RT}} = e^{\left(E_a - E_{a_2}\right)/RT}$
$2.303 \log \frac{k_2}{k_1} = \frac{E_{a_1} - E_a}{RT}$
$= \frac{(83.314-75)\times 10^3}{8.314\times 500} = 2$
$\log \mathrm{k}_2 = \frac{2}{2.303} = 0.868$
Taking Antilog $\mathrm{k}_2 = 7.38$

Q.5 Answer:

the energy below which colliding molecules will not react

Solution:

In Arrhenius equation $k = Ae^{-E_a/RT}$, E_a is the energy of activation, which is required by the colliding molecules to react resulting in the formation of products.

Q.6 Answer:

Collision frequency increases

Solution:

When the temperature is increased, energy in form of heat is supplied which increases the kinetic energy of the reacting molecules. this will increase the number of collisions and ultimately the rate of reaction will be enhanced.

Q.7 Answer:

$\ln \mathrm{A}$

Solution:

According to Arrhenius equation,
$k = A \cdot e^{-E_a/RT}$
where k **= rate constant**
or $\ln k = \ln A - \frac{E_a}{RT}$
A **= frequency factor**
E_a **= Energy of activation.**

Q.8 Answer:

$-\frac{E_a}{2.303R}$

Solution:

Arrhenius equation is given by
$k = Ae^{-E_a/(2.303RT)}$
Taking log on both sides, we get
$\log k = \log A - \frac{E_a}{2.303RT}$

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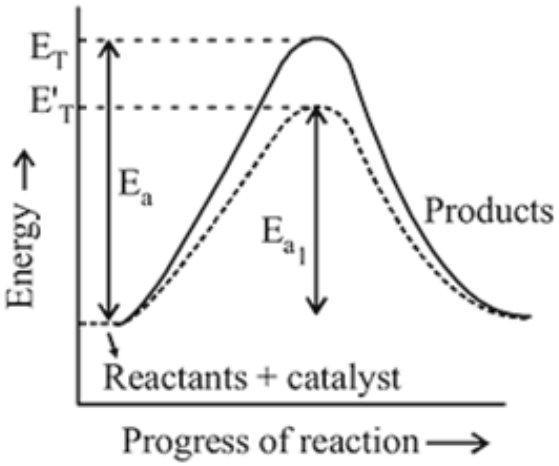
Arrhenius plot a graph between $\log k$ and $\frac{1}{T}$ whose slope is $\frac{-E_a}{2.303R}$.

Q.9 Answer:

Different from $\mathrm{E_a}$ obtained in laboratory

Solution:

The presence of enzyme (catalyst) increases the speed of reaction by lowering the energy barrier, i.e. a new path is followed with lower activation energy.



Here E_{T} is the threshold energy. $\mathrm{E_a}$ and $\mathrm{E_{a_1}}$ is energy of activation of reaction in absence and presence of catalyst respectively.

Q.10 Answer:

-20

Solution:

Presence of catalyst does not affect enthalpy change
of reaction $\Delta \mathrm{H_R} = \mathrm{E_f} - \mathrm{E_b} = 180 - 200 = -20\ \mathrm{kJ/mol}$

Q.11 Answer:

$\mathrm{A} = 1.2 \times 10^3\ \mathrm{mol^{-1}\ L\ s^{-1}}$

Solution:

$\mathrm{k} = \mathrm{Ae^{-Ea/RT}}$
$\mathrm{If T} \longrightarrow \infty, \mathrm{k} = \mathrm{A}$

Q.12 Answer:

$327^\circ \mathrm{C}$

Solution:

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$\frac{E'_a}{T_1} = \frac{E'_a}{T_2} = \frac{10}{300} = \frac{20}{T_2}$
$\therefore T_2 = 600\ K = 327^\circ C$

Q.13 Answer:

exothermic

Solution:

Exothermic because of activation energy $E_b > E_f$

Q.14 Answer:

$C > B > A$

Solution:

More activations energy, slow reaction rate.

Q.15 Answer:

less than the energy of activation of products

Solution:

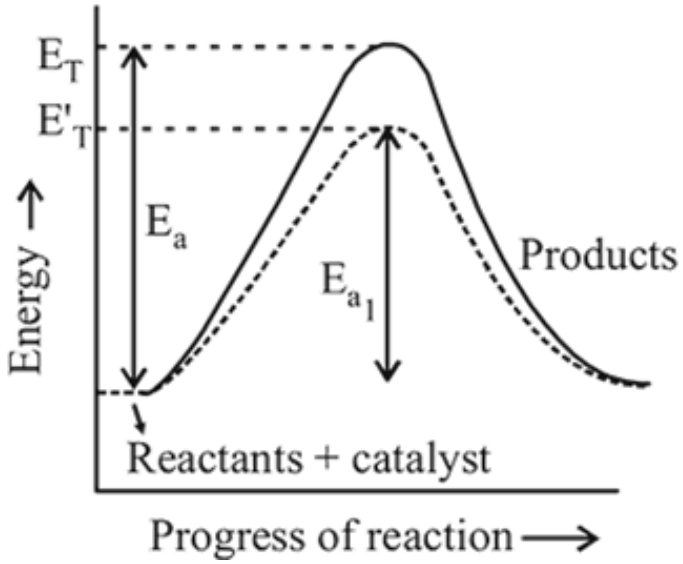
Activation energy of reactant is less than the energy of activation of products.

Q.16 Answer:

different from E_a obtained in laboratory

Solution:

The presence of enzyme (catalyst) increases the speed of reaction by lowering the energy barrier, i.e., a new path is followed with lower activation energy.



Here E_T is the threshold energy. E_a and E_{al} is energy of activation of reaction in absence and presence of catalyst respectively.

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Q.17 Answer:

6.91kcalmol^{-1}

Solution:

$e^{-E_a/RT} = 10^{-3}\% = 10^{-5}$

$E_a = 2.303 \times 2 \times 300 \times 5\text{cal}$

$= 6.91\text{kcalmol}^{-1}$

Q.18 Answer:

can be less than or more than E_a

Solution:

The activation energy of reverse reaction will depend upon whether the forward reaction is exothermic or endothermic.

As $\Delta\text{H} = E_a$ **(forward reaction)** $-E_a$ **(backward reaction)**

For exothermic reaction

$\Delta\text{H} = -\text{ve}$

$\therefore -\Delta\text{H} = E_a(\text{f}) - E_a(\text{ b})$

or $E_a(\text{f}) = E_a(\text{ b}) - \Delta\text{H}$

$\therefore E_a(\text{f}) < E_a(\text{ b})$

for endothermic reaction

$\Delta\text{H} = +\text{ve}$

$\therefore \Delta\text{H} = E_a(\text{f}) - E_a(\text{ b})$ **or** $E_a(\text{f}) = \Delta\text{H} + E_a(\text{ b})$

$\therefore E_a(\text{f}) > E_a(\text{ b})$

Q.19 Answer:

$B \rightarrow C$

Solution:

The slowest step determines the rate.

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Q.20 Answer:

$\frac{1000}{2.303}$ K

Solution:

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$k_1 = 10^{16} e^{-\frac{2000}{T}}$

$k_2 = 10^{15} e^{-\frac{1000}{T}}$

$k_1 = k_2$

$10 = e^{\frac{2000}{T} - \frac{1000}{T}}$

$\log_e 10 = \dfrac{2000}{T} - \dfrac{1000}{T}$

$\therefore \mathrm{T} = \frac{1000}{2.303}$

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